SOKOINE UNIVERSITY OF AGRICULTURE
School of Engineering and Technology

Course: ABE 721 - Agricultural Systems Modelling and Simulation
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ASSIGNMENT 5
Module 5: Simulating Materials Flow

Instructions:

- This assignment tests your ability to mathematically track the continuous, cyclical transfer of mass and energy through an agricultural ecosystem, and how to simulate human interruptions to those flows.
- Carefully read the ”Significance” statement for each question to understand the practical engineering application of the theory.
- Show all logical steps in your mathematical derivations.
- **Programming Requirement:** For questions that require computational implementation, you must submit your complete, well-commented source code (Python, MATLAB, or R) along with the exact results (console transcripts and any generated plots).

Tier 1: Simple (Conceptual Foundations)

Question 1.1: Compartment Modelling and Residence Time

Significance of Question: *Grasping the concept of residence time allows engineers to differentiate between short-term nutrient availability for crops and long-term soil carbon sequestration for climate mitigation.*

Prompt: In a steady-state ecosystem compartment model, define the concept of **Mean Residence Time (τ_i)**. If the steady-state mass of a specific microbial biomass compartment is 50 kg/ha and the measured total mass output flow is 10 kg/ha/day, what is the residence time, and what is the mathematical turnover rate of this compartment?

Question 1.2: Evaporation Paradigms

Significance of Question: *Selecting the correct evaporation paradigm prevents massive under- or over-estimations of crop water demand depending on regional wind and*

radiation conditions.

Prompt: Briefly contrast **Mass Transfer (Aerodynamic)** models with **Energy Balance** models for evaporation. In your own words, why does the **FAO-56 Penman-Monteith** equation represent a significant improvement over using either of those two models in isolation?

Tier 2: Intermediate (Application and Logic)

Question 2.1: The Advection-Dispersion-Reaction Equation (ADRE)

Significance of Question: *Understanding the separate physical mechanisms of solute transport is required to design field interventions that prevent catastrophic groundwater contamination.*

Prompt: The 1D ADRE for a dissolved material (like nitrate fertilizer) flowing through a porous soil matrix is given by:

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} - v\frac{\partial C}{\partial x} - \mu C$$

a) Define the three distinct physical or biological processes represented by the three terms on the right side of the equation.

b) When converting this PDE into a computational algorithm, why does the standard **Explicit Finite Difference Method** struggle to solve this equation when the fluid velocity (v) is extremely high? What alternative algorithm do engineers use to maintain stability?

Question 2.2: Hybrid Continuous-Discrete Simulation

Significance of Question: *Farms are not purely natural ecosystems; simulating human management requires mathematically and algorithmically interrupting biological continuity.*

Prompt: Crop growth and soil moisture depletion are governed by continuous Ordinary Differential Equations (ODEs). Explain mathematically how a simulation model handles a sudden management event, such as a farmer applying an irrigation depth of 25 mm on exactly Day 40. Use the concept of jump conditions (or Dirac Delta functions) in your explanation.

Tier 3: Hard (Synthesis and Formulation)

Question 3.1: Energy Balance and Latent Heat Formulation

Significance of Question: *Formulating the surface energy balance proves the absolute physical limit of water evaporation, ensuring irrigation models do not violate the First Law of Thermodynamics.*

Prompt: The First Law of Thermodynamics states that the net incoming solar radiation (R_n) must equal the sum of the Sensible Heat Flux (H), the Latent Heat Flux (λE), and the Soil Heat Flux (G).

a) Starting from the energy balance equation ($R_n = H + \lambda E + G$), algebraically derive the equation for the Latent Heat Flux (λE) using the Bowen Ratio ($\beta = H/\lambda E$).

b) Assume you measure a field with $R_n = 15.0$ MJ/m²/day, $G = 1.0$ MJ/m²/day, and a Bowen ratio of $\beta = 0.4$. Calculate the exact depth of water evaporated (E) in millimeters per day.
(Assume the latent heat of vaporization $\lambda \approx 2.45$ MJ/kg, and the density of water is 1000 kg/m³).

Question 3.2: Programming Compartment Flows

Significance of Question: *Computationally integrating compartment matrices translates theoretical biogeochemical cycles into predictive software tools for agronomic management.*

Prompt (Programming Task): You must write a script (in Python, MATLAB, or R) to numerically simulate a simplified 2-compartment soil carbon cycle over 100 days using the Forward Euler numerical integration method.

Let Compartment 1 be Decomposable Plant Material (DPM) and Compartment 2 be Microbial Biomass (BIO). The mass flows are defined by the following ODEs:

$$\frac{d(DPM)}{dt} = -k_1 \cdot DPM$$
$$\frac{d(BIO)}{dt} = (a \cdot k_1 \cdot DPM) - k_2 \cdot BIO$$

Given Parameters:

- Initial Conditions: $DPM_0 = 1000$ kg/ha, $BIO_0 = 100$ kg/ha.
- Decomposition rate constants: $k_1 = 0.05$ day^{-1}, $k_2 = 0.01$ day^{-1}.
- Microbial assimilation fraction: $a = 0.3$ (this means only 30% of the decomposed DPM becomes BIO; the rest is lost instantly as respired CO_2).
- Time step: $h = 1$ day. Total simulation time: 100 days.

Submission Requirements: Submit your fully commented source code, the exact console text output stating the mass of DPM and BIO at exactly Day 100, and a generated plot showing the mass of both compartments mapped over the 100-day timeline.

Please submit your work to my email kadeghe.fue@sua.ac.tz by one week after completion of Module 5. Do not send this work to kadefue@sua.ac.tz.